Teaching notes

The idea behind this resource is to explore the concept of natural selection and to explain in general why species evolve and why some become extinct. All the information to run the activity is below. The additional PowerPoint structures the activity for use on a whiteboard and includes images and links to clips.

Keywords

Adaptation, evolution, extinction, natural selection, offspring, reproduction, species, survival of the fittest

Teaching sequence

Step 1

Begin with a few images of familiar species e.g. whale, camel, fish, frog, bird

- What are they?
- How do we know?

Step 2

Then show images of animals which have unusual adaptations (ones that might be useful to the creatures described below) e.g. flying fish, giraffe, lungfish, nocturnal aye-aye; video clip of darkling beetles in Namib desert or male Namaqua sandgrouse collecting water in their feathers youtu.be/Of2Mu_D18I0 .

- Can students suggest how these adaptations arose?

Step 3

There are lots of crazy creatures in the world. Students are going to illustrate their own species of animal, **however** it must have features which enable it to live in the way described.

- A camel which lives in a very cold desert where all the trees are tall.
- A species of nocturnal frog which lives on a foggy beach.
- Fishes who live in ponds which dry up for 6 months of the year.
- A flightless, tree-climbing, fruit-eating bird.

Students work in pairs or small groups, each person in the group chooses a different creature to illustrate. They draw their animal and add notes about its adaptations (colour could be added later at home). Allow 10-15 minutes. Show and discuss the finished creatures.

Step 4

Then throw a big problem into the mix…

E.g. imagine a whale which vacuums up swarms of flying insects. What would happen if all the swarming insects suddenly vanished? If the change was too quick for the species to adapt to then it may become extinct.

What change, linked to habitat or diet, would drive their species towards extinction?

But, what if the problem was more minor?

- What if the swarming insects began to fly a little higher – out of the reach of many whales?
- How might this type of whale species alter over time?

Step 5

Students return to groups.
Possible changes are shown on the board.
Here are a few suggestions.

- For the camel the trees become taller
- For the frog the foggy spells reduce
- For the fish the ponds become drier for longer
- For the bird the fruit begins to develop at the very tips of the twigs

Students will mostly think about visible changes to their creatures' body.

Encourage them to consider physiological as well as morphological changes (depending on what they have covered so far in the syllabus).

E.g. the whale may require more energy to leap higher, may begin to have more red blood cells to carry more oxygen.

Students alter their creatures and/or add annotations.

Then pose the questions:

- Why did the trees in the camel's habitat become taller?
- Why did the fruit tree begin to produce fruit that was out of the tree-climbing birds reach?

Step 6

Show images of poison arrow frogs; poisonous mushrooms with names like 'destroying angel'; spines on an acacia; the trunk of a monkey puzzle (choose a spikey one!). They all have defences against being eaten.

The swarming insects and the trees don't want to be attacked by the students' creatures – it is not an advantage for them.

- Can they suggest adaptations which they might evolve to prevent them being eaten?

The problems which arose for the frog and the fish were not the result of a feeding relationship but the result of a change in climate.

- What would students do to protect their creature and its food source from environmental change?

Step 7

Extension questions:

- What measurements could students take over time to show the evolutionary change that has occurred in their animal species? E.g. mass of muscles used by whale to rise above the water.

- What measurements could be taken to show the main factor which has driven this adaptation? E.g. average height the insect swarm flies at.